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### Working machine including a controller configured to control an actuation valve based on traveling motor revolutions

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#### Abstract

A working machine includes a traveling pump to be operated by power of a prime mover and deliver a hydraulic fluid; a traveling motor that rotates using the delivered hydraulic fluid; an operation valve that changes pilot pressure of a pilot fluid outputted to the traveling pump in response to operation of an operation member; an actuation valve that operates based on a control signal and changes primary pressure that is the pilot pressure of the pilot fluid supplied to the operation valve; a controller that controls opening of the actuation valve by outputting the control signal to the actuation valve; and a first detector that detects an actual number of revolutions of the traveling motor. The controller includes a changer that changes setting of the control signal such that the opening of the actuation valve increases as the actual number of revolutions of the traveling motor decreases.

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## Background/Summary

### CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2022-035622 filed on Mar. 8, 2022. The entire contents of this application are hereby incorporated herein by reference.

### BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present disclosure relates to a working machine such as a skid-steer loader, a compact track loader, or the like.

## 2. Description of the Related Art

(3) A technique disclosed in Japanese Unexamined Patent Application Publication No. 2013-036274 has been known as art for preventing engine stall in a working machine such as a skid-steer loader, a compact track loader, or the like.

(4) A working machine disclosed in Japanese Unexamined Patent Application Publication No. 2013-036274 includes an engine, an HST pump configured to operate by power of the engine, a traveling manipulator for manipulating the HST pump, a pressure control valve configured to control traveling primary pressure that is pressure on a primary side of the traveling manipulator, and a controller configured to control the pressure control valve. The traveling manipulator includes a traveling lever and a pilot valve. The pilot valve outputs pilot pressure that is in proportion to an operation amount of the traveling lever.

(5) The controller performs anti-stall control for preventing the engine from stalling. In the anti-stall control, the controller prevents the engine from stalling by controlling the pressure control valve based on a light-load characteristic curve, which is used when the engine is not under significant heavy load, and a drop characteristic curve, which is used when a predetermined load or heavier is applied to the engine. In other words, the controller makes a drop in the number of revolutions of the engine as small as possible by reducing the traveling primary pressure sharply by controlling the pressure control valve when a predetermined load or heavier is applied to the working machine, thereby achieving engine stall suppression.

## SUMMARY OF THE INVENTION

(6) In the technique disclosed in Japanese Unexamined Patent Application Publication No. 2013-036274, in a middle-revolution range of the engine (prime mover), an excessive engine drop is avoided by making the output of the pressure control valve low in order to ensure a sufficient hill-climbing speed, and appropriate engine revolution (middle-revolution range) is maintained to guarantee the vehicle speed.

(7) However, when the engine is in a high-revolution range, there is a need to set the output of the pressure control valve high in order to prevent a phenomenon in which the engine revolution remains high (a phenomenon in which there is almost no engine drop because the output horsepower of the engine is in excess of the consumption horsepower of the HST pump (traveling pump)). If this phenomenon in which the engine revolution remains high occurs, it could give, to the operator, an operational feeling that the working machine is not performing work with sufficient performance (an operational feeling that the machine seems to be suppressing its horsepower output, contrary to operation by the operator).

(8) To provide a solution to the above technical issue, there is a need to set the output of the pressure control valve low when the engine is in a middle-revolution range and to set the output of the pressure control valve high when the engine is in a high-revolution range. However, if the difference between the output of the pressure control valve when the engine is in a middle-revolution range and the output of the pressure control valve when the engine is in a high-revolution range is extremely large (if the slope of the output of the pressure control valve in relation to the number of revolutions of the engine is set to be steep), there is a risk that an engine revolution control instability such as hunting of the engine might arise.

(9) Therefore, in order to give a priority to a hill-climbing speed, it is conceivable to perform output setting of the pressure control valve for a case where the engine is in a high-revolution range within a range of not triggering hunting after performing output setting of the pressure control valve for a case where the engine is in a middle-revolution range. Though this approach prevents the phenomenon in which the engine revolution remains high, it tends to result in a decrease in traveling power in a high-revolution range. Especially in work that involves opening a traveling relief valve, for example, when pushing dirt, the delivery flow rate of the HST pump (consumption

horsepower of the HST pump) decreases due to the influence of the swash-plate characteristics of the HST pump, and the engine revolution to be balanced therewith will be high, resulting in giving an operational feeling that the working machine is not performing work with sufficient performance to the operator.

(10) An object of the disclosed technique is to provide a working machine capable of giving an operational feeling that the working machine is performing work with sufficient performance to an operator.

(11) A working machine according to an aspect of the present disclosure includes: a prime mover; a traveling pump to be operated by power of the prime mover and deliver a hydraulic fluid; a traveling motor to rotate using the hydraulic fluid delivered by the traveling pump; an operation valve to change pilot pressure of a pilot fluid outputted to the traveling pump in response to operation of an operation member; an actuation valve to operate based on a control signal and change primary pressure that is the pilot pressure of the pilot fluid supplied to the operation valve; a controller configured or programmed to control opening of the actuation valve by outputting the control signal to the actuation valve; and a first detector to detect an actual number of revolutions of the traveling motor, wherein the controller includes a changer configured or programmed to change setting of the control signal such that the opening of the actuation valve increases as the actual number of revolutions of the traveling motor decreases.

(12) The working machine may further include a number-of-revolutions operation actuator, a second detector, and a storage unit. The number-of-revolutions operation actuator may be configured for an operator to operate a target number of revolutions of the prime mover. The second detector may be configured to detect an actual number of revolutions of the prime mover. A first line and a second line may be stored in the storage unit. The first line may be a line defining the control signal based on the actual number of revolutions of the prime mover in a case where a difference between the target number of revolutions of the prime mover and the actual number of revolutions of the prime mover is not less than a first threshold. The second line may be a line defining the control signal to be greater than defined by the first line in a case where the difference is less than the first threshold. The changer may be configured or programmed to change the first line by changing the control signal expressed by the first line such that the opening of the actuation valve increases as the actual number of revolutions of the traveling motor decreases.

(13) Based on a first correction coefficient defined in relation to the actual number of revolutions of the traveling motor, the changer may be configured or programmed to change the control signal expressed by the first line.

(14) At least one first function specifying a relation between the actual number of revolutions of the traveling motor and the first correction coefficient may be stored in the storage unit. The changer may be configured or programmed to calculate the first correction coefficient by substituting the actual number of revolutions of the traveling motor detected by the first detector into the at least one first function.

(15) The at least one first function may define the first correction coefficient in terms of value that is not less than one, such that the first correction coefficient corresponding to a second number of revolutions, which represents the actual number of revolutions of the traveling motor and is less than a first number of revolutions, is greater than the first correction coefficient corresponding to the first number of revolutions, which represents the actual number of revolutions of the traveling motor.

(16) The at least one first function may have degrees of inclination which are different depending on whether the actual number of revolutions of the traveling motor is less than a second threshold thereof. The degree of inclination of the at least one first function in a range where the actual number of revolutions of the traveling motor is equal to or less than the second threshold, is greater than the degree of inclination of the at least one first function in a range where the actual number of revolutions of the traveling motor is not less than the second threshold.

- (17) The controller may be operable to be switched among a plurality of modes. As the at least one first function, a plurality of first functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, may be stored in the storage unit. The changer may be configured or programmed to calculate the first correction coefficient based on, among the plurality of first functions, a first function that corresponds to a mode of the controller.
- (18) Based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer may be configured or programmed to change the control signal expressed by the first line.
- (19) At least one second function specifying a relation between the actual number of revolutions of the prime mover and the second correction coefficient may be stored in the storage unit. The changer may be configured or programmed to calculate the second correction coefficient by substituting the actual number of revolutions of the prime mover detected by the second detector into the at least one second function.
- (20) The at least one second function may define the second correction coefficient as a value equal to one when the actual number of revolutions of the prime mover is not less than a third threshold, and define the second correction coefficient in terms of value that is less than one when the actual number of revolutions of the prime mover is less than the third threshold.
- (21) The controller may be operable to be switched among a plurality of modes. As the at least one second function, a plurality of second functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, may be stored in the storage unit. The changer may be configured or programmed to calculate the second correction coefficient based on, among the plurality of second functions, a second function that corresponds to a mode of the controller.
- (22) The changer may be configured or programmed to compute a third correction coefficient based on a product of the first correction coefficient and the second correction coefficient, change the control signal expressed by the first line in a case where a value of the third correction coefficient is greater than one, and not change the control signal expressed by the first line in a case where the value of the third correction coefficient is not greater than one.
- (23) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete appreciation of preferred embodiments of the present invention and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings described below.
- (2) FIG. 1 is a diagram illustrating a hydraulic circuit for traveling in a hydraulic system (hydraulic circuit) of a working machine.
- (3) FIG. 2 is a diagram illustrating a relation between a control signal (instruction current value) and primary pressure.
- (4) FIG. 3 is a diagram illustrating an example of a setting line for a control signal (target pressure of primary pressure) based on an actual number of revolutions of a prime mover.
- (5) FIG. 4A is a diagram illustrating an example of a first function specifying a relation between an actual number of revolutions of a traveling motor and a first correction coefficient.
- (6) FIG. 4B is a diagram illustrating an example of a second function specifying a relation between the actual number of revolutions of the prime mover and a second correction coefficient.

- (7) FIG. 4C is a diagram illustrating an example of respective first functions for a plurality of modes in a second modification example.
- (8) FIG. 4D is a diagram illustrating an example of respective second functions for the plurality of modes in the second modification example.
- (9) FIG. 5A is an operation flowchart illustrating the flow of operation of a controller for changing the control signal.
- (10) FIG. 5B is an operation flowchart illustrating the flow of operation of the controller for changing the control signal according to a first modification example.
- (11) FIG. 5C is an operation flowchart illustrating the flow of operation of the controller for changing the control signal according to the second modification example.
- (12) FIG. 6 is a diagram illustrating an example of a display according to the second modification example, and a switching screen displayed on a display unit thereof.
- (13) FIG. 7 is a diagram for explaining a mode display portion displayed by the display and the display unit according to the second modification example.
- (14) FIG. 8 is a diagram illustrating an example of a case where an actuation valve is provided on a secondary side of an operation valve according to a third modification example.
- (15) FIG. 9 is a diagram illustrating a modified configuration according to a fourth modification example in which a hydraulic-type manipulator is replaced with an electric-type manipulator such as a joystick.

(16) FIG. 10 is a side view of a track loader that is an example of the working machine.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (17) The preferred embodiments will now be described with reference to the accompanying drawings, wherein like reference numerals designate corresponding or identical elements throughout the various drawings. The drawings are to be viewed in an orientation in which the reference numerals are viewed correctly.
- (18) With reference to the accompanying drawings, an exemplary embodiment of the disclosed technique will now be explained.
- (19) FIG. 10 is a side view of a track loader that is an example of a working machine 1. In FIG. 10, a compact track loader is illustrated as an example of the working machine 1. However, the working machine 1 according to the present disclosure is not limited to a compact track loader but may be any other kind of loader machine such as, for example, a skid-steer loader. The working machine 1 may be any kind of working machine other than a loader machine.
- (20) As illustrated in FIG. 10, the working machine 1 includes a machine body 2, a cabin 3, a working device 4, and a traveling device 5. In an exemplary embodiment of the present disclosure, the term “forward” will be used for referring to a direction which an operator seated on an operator's seat 8 of the working machine 1 faces (leftward in FIG. 10), and the term “rearward” will be used for referring to the opposite direction thereof (rightward in FIG. 10). The term “leftward” will be used for referring to a direction going toward the left side as viewed from the operator (direction toward the near side in FIG. 10), and the term “rightward” will be used for referring to a direction going toward the right side as viewed from the operator (direction toward the far side in FIG. 10). A horizontal direction orthogonal to a front-rear direction will be referred to as “machine-body width direction”.
- (21) The cabin 3 is mounted on the machine body 2. The operator's seat 8 is provided inside the cabin 3. The working device 4 is mounted on the machine body 2. A prime mover 6 is mounted on a rear portion inside the machine body 2. The traveling device 5 is provided outside the machine body 2. The traveling device 5 includes a first traveling device 5L provided on the left side of the machine body 2 and a second traveling device 5R provided on the right side of the machine body 2.
- (22) With reference to FIG. 10, the working device 4 will now be described in detail. The working device 4 includes booms 10, a working tool 11, lift links 12, control links 13, boom cylinders 14, and bucket cylinders 15.

(23) The booms **10** are provided to the left and right of the cabin **3** respectively such that they can be moved pivotally up and down. The working tool **11** is, for example, a bucket. The working tool **11** is provided on a first end portion (front end portion) **10a** of the booms **10** such that it can be moved pivotally up and down. The lift links **12** and the control links **13** support a second end portion (rear end portion) **10b** of the booms **10**, which is the opposite of the first end portion **10a** thereof, to enable pivotal up-and-down motion of the booms **10**. The boom cylinders **14** raise and lower the booms **10** by their extending-and-retracting motion. The bucket cylinders **15** move the working tool **11** pivotally by their extending-and-retracting motion.

(24) The first end portion (front end portion) **10a** of the left boom **10** and the first end portion (front end portion) **10a** of the right boom **10** are coupled to each other via a non-standard-shaped coupling pipe (not illustrated). The second end portion (rear end portion) **10b** of the left boom **10** and the second end portion (rear end portion) **10b** of the right boom **10** are coupled to each other via a round coupling pipe.

(25) The lift links **12**, the control links **13**, and the boom cylinders **14** are provided on the left and right sides with respect to the machine body **2** respectively for the left and right booms **10**.

(26) The lift link **12** is provided in vertical orientation behind the second end portion **10b** of each of the booms **10**. A first end portion (upper end portion) **12a** of the lift links **12** is pivotally supported on a pivot shaft **16** near the rear end of the second end portion **10b** of the booms **10** in such a way as to be able to rotate on its horizontal axis. A second end portion (lower end portion) **12b** of the lift links **12**, which is the opposite of the first end portion **12a** thereof, is pivotally supported on a pivot shaft **17** near the rear end of the machine body **2** in such a way as to be able to rotate on its horizontal axis.

(27) A first end portion (upper end portion) **14a** of the boom cylinders **14** is pivotally supported on a pivot shaft **18** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **18** is provided near a front portion of the second end portion **10b** of the booms **10**. A second end portion (lower end portion) **14b** of the boom cylinders **14**, which is the opposite of the first end portion **14a** thereof, is pivotally supported on a pivot shaft **19** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **19** is provided in the lower rear portion of the machine body **2**.

(28) The control links **13** are provided in front of the lift links **12**. A first end portion (front end portion) **13a** of the control links **13** is pivotally supported on a pivot shaft **20** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **20** is provided on the machine body **2** in front of the lift links **12**. A second end portion (rear end portion) **13b** of the control links **13**, which is the opposite of the first end portion **13a** thereof, is pivotally supported on a pivot shaft **21** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **21** is provided on the booms **10** ahead of, and above, the pivot shaft **17**.

(29) Therefore, the booms **10** move pivotally up and down around the pivot shaft **16** due to the extending-and-retracting motion of the boom cylinders **14**, with their second end portion **10b** supported by the lift links **12** and the control links **13**. This pivotal motion raises and lowers the first end portion **10a** of the booms **10**. The control links **13** move pivotally up and down around the pivot shaft **20** when the booms **10** move pivotally up and down. The lift links **12** move pivotally forward and rearward around the pivot shaft **17** when the control links **13** move pivotally up and down.

(30) In FIG. **10**, a bucket is attached as an example of the working tool **11** to the first end portion **10a** of the booms **10**. However, an alternative working tool **11** can be attached to the first end portion **10a** of the booms **10** in place of the bucket. The alternative working tool **11** attachable to the first end portion **10a** of the booms **10** is an attachment (auxiliary attachment) such as, for example, a hydraulic crusher, a hydraulic breaker, an angle broom, an earth auger, a pallet fork, a sweeper, a mower, or a snow blower. The auxiliary attachment includes hydraulic devices such as a hydraulic motor and a hydraulic cylinder and operates using a hydraulic fluid supplied to it.

(31) A connection member **25** is provided on the first end portion **10a** of the left boom **10**. The

connection member **25** is a member for connection between a first conduit member (not illustrated) connected to the auxiliary attachment and a second conduit member (not illustrated) such as a pipe provided on the boom **10**. Specifically, the first conduit member connected to the auxiliary attachment is connected to a first end portion (front end portion) **25a** of the connection member **25**. The second conduit member is connected to a second end portion (rear end portion) **25b**, which is the opposite of the first end portion **25a**. The connection enables a hydraulic fluid flowing through the second conduit member to be supplied to the auxiliary attachment through the first conduit member.

(32) The bucket cylinders **15** are disposed near the first end portion **10a** of the booms **10** respectively. A first end portion (upper end portion) **15a** of the bucket cylinders **15** is pivotally supported on a pivot shaft **22** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **22** is provided near the rear end of the first end portion **10a** of the booms **10**. A second end portion (lower end portion) **15b** of the bucket cylinders **15**, which is the opposite of the first end portion **15a** thereof, is pivotally supported on a pivot shaft **23** in such a way as to be able to rotate on its horizontal axis. The pivot shaft **23** is provided on an upper rear portion of the working tool **11**. Having this structure, the bucket cylinders **15** move the bucket **11** pivotally by their extending-and-retracting motion.

(33) In the present embodiment, a crawler-type traveling system is adopted for each of the traveling device **5** on the left side (first traveling device **5L**) and the traveling device **5** on the right side (second traveling device **5R**). The traveling device **5** is not limited to a crawler-type traveling device such as one illustrated in FIG. **10**. The traveling device **5** may be a semi-crawler-type traveling device or a wheeled-type traveling device having front and rear wheels.

(34) The prime mover **6** is an internal combustion engine (engine) such as a diesel engine or a gasoline engine, or an electric motor, etc. In the present embodiment, the prime mover **6** is a diesel engine, but is not limited thereto.

(35) With reference to FIG. **1**, a hydraulic system for traveling will now be explained. FIG. **1** is a diagram illustrating a hydraulic circuit for traveling in a hydraulic system (hydraulic circuit) of the working machine **1**. The hydraulic system for traveling of the working machine **1** is a system for causing the traveling device **5** to operate. As illustrated in FIG. **1**, the working machine **1** includes a controller **100**, a first hydraulic pump **P1**, a second hydraulic pump **P2**, a traveling pump **50**, and a traveling motor **51**. The controller **100** is a device configured in the form of an electric/electronic circuit, a program stored in a CPU or an MPU, or the like. The controller **100** controls various devices of the working machine **1**. The controller **100** has a storage unit **100a**. The storage unit **100a** is a nonvolatile memory or the like. Various kinds of information regarding control by the controller **100** and the like are stored in the storage unit **100a**.

(36) The first hydraulic pump **P1** operates by power of the prime mover **6** and delivers a hydraulic fluid. The first hydraulic pump **P1** is a constant-displacement-type gear pump. Specifically, the first hydraulic pump **P1** is connected between a hydraulic fluid tank **T** and a delivery fluid passage **40** and is capable of delivering a hydraulic fluid contained in the hydraulic fluid tank **T** to the delivery fluid passage **40**. More particularly, the first hydraulic pump **P1** delivers a hydraulic fluid to be used mainly for control of the working machine **1**.

(37) In the description below, of the hydraulic fluid delivered from the first hydraulic pump **P1**, the hydraulic fluid to be used for control may be referred to as “pilot fluid”, and the pressure of the pilot fluid may be referred to as “pilot pressure”.

(38) The second hydraulic pump **P2** operates by power of the prime mover **6** and delivers a hydraulic fluid. The second hydraulic pump **P2** is a constant-displacement-type gear pump. The second hydraulic pump **P2** is connected between the hydraulic fluid tank **T** and a main fluid passage **45** and is capable of delivering a hydraulic fluid contained in the hydraulic fluid tank **T** to the main fluid passage **45**. More particularly, the second hydraulic pump **P2** delivers a hydraulic fluid to a hydraulic system for work, which is not for traveling.



(39) The traveling pump **50** and the traveling motor **51** are hydraulic-fluid-driven devices. In the description below, the traveling pump **50** and the traveling motor **51** may be referred to as hydraulic devices **S**. The traveling pump **50** is a pump configured to operate by power of the prime mover **6**. In the present embodiment, the traveling pump **50** includes a first traveling pump **SOL** and a second traveling pump **50R**. Specifically, the traveling pump **50** is a swash-plate variable displacement axial pump configured to operate by power of the prime mover **6**. The traveling pump **50** includes a forward traveling pressure receiver **50a** and a rearward traveling pressure receiver **50b** on which pilot pressure acts. An angle of a swash plate of the traveling pump **50** changes according to the pilot pressure acting on the forward traveling pressure receiver **50a** and the rearward traveling pressure receiver **50b**.

(40) The traveling pump **50** is capable of changing a delivery amount (output) of a hydraulic fluid supplied from the delivery fluid passage **40** and changing a delivery direction of the hydraulic fluid by the change in the angle of the swash plate.

(41) The traveling motor **51** is a motor configured to operate using the hydraulic fluid delivered from the traveling pump **50** and transmit power to the drive shaft of the traveling device **5**. In the present embodiment, the traveling motor **51** includes a first traveling motor **51L** and a second traveling motor **51R**.

(42) The first traveling motor **51L** is a motor configured to transmit power to the drive shaft of the traveling device **5** (first traveling device **5L**) provided on the left side of the machine body **2**. The first traveling motor **51L** is capable of operating using the hydraulic fluid delivered from the first traveling pump **SOL**. Specifically, the first traveling motor **51L** is connected to the first traveling pump **SOL** through a circulation fluid passage **53a**. Therefore, the first traveling pump **SOL** is capable of delivering the hydraulic fluid to the first traveling motor **51L** through the circulation fluid passage **53a**.

(43) The first traveling motor **51L** is capable of changing its rotation speed (number of revolutions) based on the flow rate of the hydraulic fluid supplied from the first traveling pump **SOL**.

(44) The first traveling motor **51L** is capable of changing its rotation speed to a first speed (a predetermined low-speed range), which is **LOW**, and to a second speed (a predetermined high-speed range), which is **HIGH**. In the present embodiment, the rotation speed (number of revolutions) of the first traveling motor **51L** can be changed by the extending-and-retracting motion of a swash plate switching cylinder **52L**. Specifically, as illustrated in FIG. **1**, the swash plate switching cylinder **52L** is connected to the first traveling motor **51L**. The rotation speed of the first traveling motor **51L** is set to the first speed when the swash plate switching cylinder **52L** is retracted. The rotation speed of the first traveling motor **51L** is set to the second speed when the swash plate switching cylinder **52L** is extended.

(45) The second traveling motor **51R** is a motor configured to transmit power to the drive shaft of the traveling device **5** (second traveling device **5R**) provided on the right side of the machine body **2**. The second traveling motor **51R** is capable of operating using the hydraulic fluid delivered from the second traveling pump **50R**. Specifically, the second traveling motor **51R** is connected to the second traveling pump **50R** through a circulation fluid passage **53b**. Therefore, the second traveling pump **50R** is capable of delivering the hydraulic fluid to the second traveling motor **51R** through the circulation fluid passage **53b**.

(46) The second traveling motor **51R** is capable of changing its rotation speed (number of revolutions) based on the flow rate of the hydraulic fluid supplied from the second traveling pump **50R**.

(47) The second traveling motor **51R** is capable of changing its rotation speed to a first speed (a predetermined low-speed range), which is **LOW**, and to a second speed (a predetermined high-speed range), which is **HIGH**. In the present embodiment, the rotation speed (number of revolutions) of the second traveling motor **51R** can be changed by the extending-and-retracting motion of a swash plate switching cylinder **52R**. Specifically, as illustrated in FIG. **1**, the swash

plate switching cylinder **52R** is connected to the second traveling motor **51R**. The rotation speed of the second traveling motor **51R** is set to the first speed when the swash plate switching cylinder **52R** is retracted. The rotation speed of the second traveling motor **51R** is set to the second speed when the swash plate switching cylinder **52R** is extended.

(48) As illustrated in FIG. **1**, a traveling relief valve **71** is provided on the circulation fluid passage **53a**, through which the first traveling motor **51L** is connected to the first traveling pump SOL, and another traveling relief valve **71** is provided on the circulation fluid passage **53b**, through which the second traveling motor **51R** is connected to the second traveling pump **50R**. The traveling relief valve **71** drains, to the hydraulic fluid tank T, the hydraulic fluid flowing through the circulation fluid passage **53a**, **53b** when the working machine **1** performs work such as pushing dirt by using its bucket.

(49) Next, operation regarding the traveling of the working machine **1**, that is, operation of the traveling device **5** (traveling operation), will now be explained in detail. As illustrated in FIG. **1**, the working machine **1** includes a traveling manipulator (operation device) **54**.

(50) The manipulator **54** is a device for manipulating the traveling pump **50** (the first traveling pump **50L** and the second traveling pump **50R**). The manipulator **54** is capable of changing the angle of the swash plate (swash-plate angle) of the traveling pump **50** by changing the pilot pressure acting on the forward traveling pressure receiver **50a** and the rearward traveling pressure receiver **50b**. The manipulator **54** includes an operation member (a traveling lever) **55** and a plurality of operation valves (traveling operation valves) **56**.

(51) The operation member **55** is an operation lever configured to be swung in the left-right direction (machine-body width direction) or the front-rear direction selectively. The operation member **55** is supported on the operation valves **56**. When the neutral position N of the operation member **55** is defined as its home position, the operation member **55** can be operated forward (in the direction indicated by an arrow A1 in FIG. **1**) and rearward (in the direction indicated by an arrow A2 in FIG. **1**) from the neutral position N, and leftward (in the direction indicated by an arrow A3 in FIG. **1**) and rightward (in the direction indicated by an arrow A4 in FIG. **1**) from the neutral position N.

(52) In other words, the operation member **55** can be swung in at least four directions from the neutral position N defined as its home position. For the sake of explanation, in the description of the operation member **55** below, the two directions going forward and rearward, namely, the front-rear direction, may be hereinafter referred to as “first direction”. The two directions going leftward and rightward, namely, the left-right direction (machine-body width direction), may be hereinafter referred to as “second direction”.

(53) Each of the plurality of operation valves **56** is a valve configured to be manipulated by operating the operation member **55**. Specifically, the operation valves **56** are connected to the delivery fluid passage **40** and are capable of changing the pressure of the pilot fluid (pilot pressure) that is the hydraulic fluid supplied from the delivery fluid passage **40**. The operation valves **56** are manipulated by operating the operation member **55** that is common to them, that is, a single operation lever. The operation valves **56** are a first pilot valve **56A**, a second pilot valve **56B**, a third pilot valve **56C**, and a fourth pilot valve **56D**.

(54) When the operation member **55** is swung forward as one of the front-rear direction (first direction), that is, when the operation member **55** is operated forward, the pressure of the pilot fluid which the first pilot valve **56A** outputs changes in accordance with an amount of the forward operation. When the operation member **55** is swung rearward as the other of the front-rear direction (first direction), that is, when the operation member **55** is operated rearward, the pressure of the pilot fluid which the second pilot valve **56B** outputs changes in accordance with an amount of the rearward operation.

(55) When the operation member **55** is swung leftward as one of the left-right direction (second direction), that is, when the operation member **55** is operated leftward, the pressure of the pilot

fluid which the third pilot valve **56C** outputs changes in accordance with an amount of the leftward operation. When the operation member **55** is swung rightward as the other of the left-right direction (second direction), that is, when the operation member **55** is operated rightward, the pressure of the pilot fluid which the fourth pilot valve **56D** outputs changes in accordance with an amount of the rightward operation.

(56) As illustrated in FIG. **1**, the plurality of operation valves **56** is connected to the traveling pump **50** through a traveling fluid passage **42**. That is, the traveling pump **50** is a hydraulic device that is operable using the pilot fluid outputted from the operation valve **56** (the first pilot valve **56A**, the second pilot valve **56B**, the third pilot valve **56C**, and the fourth pilot valve **56D**).

(57) As illustrated in FIG. **1**, the traveling fluid passage **42** is a fluid passage for connection between the plurality of operation valves **56** and the traveling pump **50**. The traveling fluid passage **42** includes a first traveling fluid passage **42a**, a second traveling fluid passage **42b**, a third traveling fluid passage **42c**, a fourth traveling fluid passage **42d**, and a fifth traveling fluid passage **42e**.

(58) The first traveling fluid passage **42a** is a fluid passage connected to the forward traveling pressure receiver **50a** of the first traveling pump SOL. The second traveling fluid passage **42b** is a fluid passage connected to the rearward traveling pressure receiver **50b** of the first traveling pump **50L**. The third traveling fluid passage **42c** is a fluid passage connected to the forward traveling pressure receiver **50a** of the second traveling pump **50R**. The fourth traveling fluid passage **42d** is a fluid passage connected to the rearward traveling pressure receiver **50b** of the second traveling pump **50R**. The fifth traveling fluid passage **42e** is a fluid passage for connection of the plurality of operation valves **56** to the first traveling fluid passage **42a**, the second traveling fluid passage **42b**, the third traveling fluid passage **42c**, and the fourth traveling fluid passage **42d**. More specifically, the fifth traveling fluid passage **42e** includes a bridge portion **42e1**, which is connected to the first traveling fluid passage **42a**, the second traveling fluid passage **42b**, the third traveling fluid passage **42c**, and the fourth traveling fluid passage **42d** and has a plurality of shuttle valves **43**, and a coupling passage **42e2**, which is a passage for connection between a junction of the bridge portion **42e1** and the plurality of operation valves **56**.

(59) When the operation member **55** is swung forward (in the direction indicated by the arrow **A1** in FIG. **1**), the first pilot valve **56A** is manipulated, and the first pilot valve **56A** outputs a pilot fluid. The pressure of the pilot fluid outputted from the first pilot valve **56A** (pilot pressure) acts on the forward traveling pressure receiver **50a** of the first traveling pump SOL through the fifth traveling fluid passage **42e** and the first traveling fluid passage **42a**. In addition, the pressure of the pilot fluid outputted from the first pilot valve **56A** (pilot pressure) acts on the forward traveling pressure receiver **50a** of the second traveling pump SOR through the fifth traveling fluid passage **42e** and the third traveling fluid passage **42c**. This changes the swash-plate angles of the first traveling pump SOL and the second traveling pump SOR to cause the first traveling motor **51L** and the second traveling motor **51R** to operate in the normal direction of rotation (toward the forward rotation side), thereby causing the working machine **1** to travel straight forward.

(60) When the operation member **55** is swung rearward (in the direction indicated by the arrow **A2** in FIG. **1**), the second pilot valve **56B** is manipulated, and the second pilot valve **56B** outputs a pilot fluid. The pressure of the pilot fluid outputted from the second pilot valve **56B** (pilot pressure) acts on the rearward traveling pressure receiver **50b** of the first traveling pump **50L** through the fifth traveling fluid passage **42e** and the second traveling fluid passage **42b**. In addition, the pressure of the pilot fluid outputted from the second pilot valve **56B** (pilot pressure) acts on the rearward traveling pressure receiver **50b** of the second traveling pump **50R** through the fifth traveling fluid passage **42e** and the fourth traveling fluid passage **42d**. This changes the swash-plate angles of the first traveling pump **50L** and the second traveling pump SOR to cause the first traveling motor **51L** and the second traveling motor **51R** to operate in the reverse direction of rotation (toward the rearward rotation side), thereby causing the working machine **1** to travel

straight rearward.

(61) When the operation member **55** is swung leftward (in the direction indicated by the arrow **A3** in FIG. **1**), the third pilot valve **56C** is manipulated, and the third pilot valve **56C** outputs a pilot fluid. The pressure of the pilot fluid outputted from the third pilot valve **56C** (pilot pressure) acts on the forward traveling pressure receiver **50a** of the second traveling pump **SOR** through the fifth traveling fluid passage **42e** and the third traveling fluid passage **42c**. In addition, the pressure of the pilot fluid outputted from the third pilot valve **56C** (pilot pressure) acts on the rearward traveling pressure receiver **50b** of the first traveling pump **SOL** through the fifth traveling fluid passage **42e** and the second traveling fluid passage **42b**. This changes the swash-plate angles of the first traveling pump **SOL** and the second traveling pump **SOR** to cause the first traveling motor **51L** to operate in the reverse direction of rotation and the second traveling motor **51R** to operate in the normal direction of rotation, thereby causing the working machine **1** to make a turn to the left.

(62) When the operation member **55** is swung rightward (in the direction indicated by the arrow **A4** in FIG. **1**), the fourth pilot valve **56D** is manipulated, and the fourth pilot valve **56D** outputs a pilot fluid. The pressure of the pilot fluid outputted from the fourth pilot valve **56D** (pilot pressure) acts on the forward traveling pressure receiver **50a** of the first traveling pump **SOL** through the fifth traveling fluid passage **42e** and the first traveling fluid passage **42a**. In addition, the pressure of the pilot fluid outputted from the fourth pilot valve **56D** (pilot pressure) acts on the rearward traveling pressure receiver **50b** of the second traveling pump **SOR** through the fifth traveling fluid passage **42e** and the fourth traveling fluid passage **42d**. This changes the swash-plate angles of the first traveling pump **50L** and the second traveling pump **50R** to cause the first traveling motor **51L** to operate in the normal direction of rotation and the second traveling motor **51R** to operate in the reverse direction of rotation, thereby causing the working machine **1** to make a turn to the right.

(63) When the operation member **55** is swung in an oblique direction, the rotation direction and the rotation speed of each of the first traveling motor **51L** and the second traveling motor **51R** are determined based on differential pressure between the pilot pressure acting on the forward traveling pressure receiver **50a** and the pilot pressure acting on the rearward traveling pressure receiver **50b**, and the working machine **1** makes a turn to the right or a turn to the left while traveling forward or rearward.

(64) More specifically, when the operation member **55** is swung obliquely forward to the left, the working machine **1** makes a turn to the left while traveling forward at a speed corresponding to the swing angle of the operation member **55**. When the operation member **55** is swung obliquely forward to the right, the working machine **1** makes a turn to the right while traveling forward at a speed corresponding to the swing angle of the operation member **55**. When the operation member **55** is swung obliquely rearward to the left, the working machine **1** makes a turn to the left while traveling rearward at a speed corresponding to the swing angle of the operation member **55**. When the operation member **55** is swung obliquely rearward to the right, the working machine **1** makes a turn to the right while traveling rearward at a speed corresponding to the swing angle of the operation member **55**.

(65) As illustrated in FIG. **1**, the working machine **1** includes a traveling switching valve **57**. The traveling switching valve **57** is switchable between a first state, in which the rotation speed (number of revolutions) of the traveling motor **51** is set to the first speed, and a second state, in which the rotation speed (number of revolutions) of the traveling motor **51** is set to the second speed. The traveling switching valve **57** includes first switching valves **58L** and **58R** and a second switching valve **59**.

(66) The first switching valve **58L** is a two-position switching valve switchable between a first position **58L1** and a second position **58L2**. The first switching valve **58L** is connected to the swash plate switching cylinder **52L** through a fluid passage. The first switching valve **58L**, when put into the first position **58L1**, stops hydraulic fluid supply to the swash plate switching cylinder **52L**, thereby making the swash plate switching cylinder **52L** retracted. The first switching valve **58L**,

when put into the second position **58L2**, supplies a hydraulic fluid to the swash plate switching cylinder **52L**, thereby making the swash plate switching cylinder **52L** extended.

(67) The first switching valve **58R** is a two-position switching valve switchable between a first position **58R1** and a second position **58R2**. The first switching valve **58R** is connected to the swash plate switching cylinder **52R** through a fluid passage. The first switching valve **58R**, when put into the first position **58R1**, stops hydraulic fluid supply to the swash plate switching cylinder **52R**, thereby making the swash plate switching cylinder **52R** retracted. The first switching valve **58R**, when put into the second position **58R2**, supplies a hydraulic fluid to the swash plate switching cylinder **52R**, thereby making the swash plate switching cylinder **52R** extended.

(68) The second switching valve **59** is a solenoid valve configured to switch the position of the first switching valve **58L** and the position of the first switching valve **58R**. Specifically, the second switching valve **59** is a two-position switching valve switchable between a first position **59a** and a second position **59b** by energization. As illustrated in FIG. 1, the second switching valve **59** is connected to the first switching valves **58L** and **58R** through a fluid passage **41**. The second switching valve **59**, when put into the first position **59a**, supplies a hydraulic fluid to a pressure receiver of the first switching valve **58L** and a pressure receiver of the first switching valve **58R** to switch the first switching valve **58L** to the first position **58L1** and switch the first switching valve **58R** to the first position **58R1**. The second switching valve **59**, when put into the second position **59b**, stops the supply of the hydraulic fluid to the pressure receiver of the first switching valve **58L** and the pressure receiver of the first switching valve **58R** to switch the first switching valve **58L** to the second position **58L2** and switch the first switching valve **58R** to the second position **58R2**.

(69) That is, when the second switching valve **59** is put into the first position **59a**, the traveling switching valve **57** is put into the first state, the swash plate switching cylinders **52L** and **52R** are retracted, and the rotation speed of the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) is switched to the first speed. When the second switching valve **59** is put into the second position **59b**, the traveling switching valve **57** is put into the second state, the swash plate switching cylinders **52L** and **52R** are extended, and the rotation speed of the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) is switched to the second speed.

(70) Therefore, the traveling switching valve **57** is capable of switching the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) between the first speed and the second speed.

(71) As illustrated in FIG. 1, the working machine **1** includes a switching actuator (speed operation actuator) **101** operable by the operator or the like. The switching actuator **101** is a selector switch to be operated for switching the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) between the first speed and the second speed. The switching actuator **101** is connected to the controller **100**, is operated by the operator or the like, and inputs an operation signal into the controller **100**. Based on switching operation of the switching actuator **101**, the controller **100** outputs a control signal (for example, a voltage, a current, etc.) to the traveling switching valve **57** to switch the traveling switching valve **57** between the first state and the second state.

(72) In this way, the switching actuator (selector switch) **101** can be operated for switching the traveling switching valve **57** from the first state to the second state and for switching the traveling switching valve **57** from the second state to the first state. That is, the switching actuator **101** can be operated for shift-up switching of switching the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) from the first speed (first state) to the second speed (second state) and for shift-down switching of switching the traveling motor **51** (the first traveling motor **51L** and the second traveling motor **51R**) from the second speed (second state) to the first speed (first state).

(73) The working machine **1** according to the present disclosure is capable of reducing the output

of the hydraulic devices S (the traveling pump 50 and the traveling motor 51) and thereby suppressing engine stall. Specifically, the working machine 1 suppresses engine stall by reducing the output of the hydraulic devices S by changing the pilot pressure (primary pressure) of the pilot fluid supplied to the operation valves 56. Anti-stall control will now be explained in detail.

(74) As illustrated in FIG. 1, the working machine 1 includes a number-of-revolutions operation actuator (accelerator) 102 and a second detector 103. The number-of-revolutions operation actuator 102 is a member for operating a target number of revolutions of the prime mover 6. The number-of-revolutions operation actuator 102 is connected to the controller 100 and inputs an operation signal into the controller 100. The number-of-revolutions operation actuator 102 is provided near the operator's seat 8. The number-of-revolutions operation actuator 102 is an accelerator lever supported pivotally, an accelerator pedal supported pivotally, an accelerator potentiometer supported rotatably, an accelerator slider supported slidably, or the like. The number-of-revolutions operation actuator 102 is not limited to these examples. Any member, etc. may be adopted as long as it can be used for operating a target number of revolutions of the prime mover 6.

(75) The second detector 103 is, for example, a sensor configured to detect an actual number of revolutions of the prime mover 6. The second detector 103 is connected to the controller 100 and inputs a signal that is based on the detection (a detection signal) into the controller 100.

(76) Based on the difference between the target number of revolutions of the prime mover 6 operated using the number-of-revolutions operation actuator 102 and the actual number of revolutions of the prime mover 6 detected by the second detector 103 (referred to as “drop in number of revolutions”), the controller 100 performs control for suppressing the stopping of the prime mover 6, that is, control for suppressing engine stall (anti-stall control). In the present embodiment, in anti-stall control, the controller 100 suppresses engine stall by reducing the output of the traveling pump 50 in a case where the drop in number of revolutions is not less than a first threshold.

(77) As illustrated in FIG. 1, the working machine 1 includes an actuation valve 70. The actuation valve 70 is a valve capable of changing the pilot pressure of the pilot fluid for actuating the traveling pump 50. The actuation valve 70 is provided on the delivery fluid passage 40 and changes the pilot pressure (primary pressure) of the pilot fluid supplied from the delivery fluid passage 40 to the manipulator 54 (the plurality of operation valves 56).

(78) The actuation valve 70 operates based on a control signal (for example, a voltage, a current, etc.) of the controller 100 and, by changing its opening, changes the pilot pressure (primary pressure) of the pilot fluid supplied from the delivery fluid passage 40 to the manipulator 54. A case where the control signal of the controller 100 is a current will be described below. In the description below, the value of the current outputted as the control signal of the controller 100 will be referred to as “instruction current value”. A proportional solenoid valve configured as the actuation valve 70 is capable of increasing its opening in proportion to the instruction current value.

(79) That is, the primary pressure changes according to the control signal outputted from the controller 100 to the actuation valve 70. In the present embodiment, since the controller 100 outputs the instruction current value to the actuation valve 70, the primary pressure changes according to the instruction current value outputted from the controller 100 to the actuation valve 70. Specifically, when the instruction current value outputted by the controller 100 to the actuation valve 70 increases, the opening of the actuation valve 70 increases, and the primary pressure increases. When the instruction current value outputted by the controller 100 to the actuation valve 70 decreases, the opening of the actuation valve 70 decreases, and the primary pressure decreases.

(80) FIG. 2 is a diagram illustrating a relation between the control signal (instruction current value) and the primary pressure. As illustrated in FIG. 2, there is a proportional relation, or a correspondence close to a proportional relation (correlation), between the instruction current value and the primary pressure. Therefore, by changing the instruction current value that is the value of the control signal outputted to the actuation valve 70, the controller 100 is capable of changing the

target pressure of the pilot pressure (primary pressure) of the pilot fluid supplied to the plurality of operation valves **56**.

(81) As illustrated in FIG. **1**, the controller **100** includes a processor **100b**, which is configured in the form of an electric/electronic circuit provided in the controller **100**, a program stored in a CPU, or the like.

(82) The processor **100b** computes the drop in number of revolutions by subtracting the actual number of revolutions of the prime mover **6** detected by the second detector **103** from the target number of revolutions of the prime mover **6** operated using the number-of-revolutions operation actuator **102**. In addition, based on the actual number of revolutions of the prime mover **6** detected by the second detector **103** and the computed drop in number of revolutions, the processor **100b** defines the control signal (instruction current value) outputted to the actuation valve **70**. The processor **100b** acquires a control map pre-stored in the storage unit **100a** and refers to a setting line L prescribed in the control map.

(83) A control map containing the setting line L is stored in the storage unit **100a**. FIG. **3** is a diagram illustrating an example of the setting line L for the control signal (the target pressure of the primary pressure) based on the actual number of revolutions of the prime mover **6**. The setting line L is a function for the processor **100b** (controller **100**) to define the control signal, based on the actual number of revolutions of the prime mover **6** detected by the second detector **103**. The setting line L is prescribed based on a relation between the instruction current value and the actual number of revolutions of the prime mover **6** when the operation valve **56** is fully open. In the example illustrated in FIG. **3**, a control map showing an example of the setting line L (anti-stall map) is depicted. The setting line L includes a first line La and a second line Lb.

(84) As has been described above, there is a correspondence such as one close to a proportional relation between the instruction current value that is the value of the control signal outputted by the controller **100** to the actuation valve **70** and the target pressure of the primary pressure (see FIG. **2**). That is, the setting line L (the first line La and the second line Lb) illustrated in FIG. **3** can be paraphrased as a setting line L for defining the target pressure of the primary pressure corresponding to the control signal (instruction current value) based on the actual number of revolutions of the prime mover **6**. Therefore, the vertical axis of FIG. **3** may be labeled as “control signal (instruction current value)” or as “primary pressure (target pressure)”.

(85) The first line La is a curve for, based on the actual number of revolutions, setting the control signal (instruction current value) corresponding to the target pressure of the primary pressure in a case where the drop in number of revolutions computed by the processor **100b** is not less than the first threshold. That is, the first line La is a drop characteristic curve to be used when a traveling load not lighter than a predetermined load is applied to the prime mover **6**.

(86) The second line Lb is a curve for, based on the actual number of revolutions, setting the control signal (instruction current value) corresponding to the target pressure of the primary pressure in a case where the drop in number of revolutions computed by the processor **100b** is less than the first threshold. The second line Lb is a curve for setting the value of the control signal (instruction current value) to be larger than the value set using the first line La. That is, the second line Lb is a light-load characteristic curve to be used when the load applied to the prime mover **6** is lighter than the predetermined load.

(87) As illustrated in FIG. **1**, the controller **100** includes a changer **100c**. The changer **100c** is configured in the form of an electric/electronic circuit provided in the controller **100**, a program stored in a CPU, or the like. The changer **100c** changes (corrects) the control signal (instruction current value) computed using the first line La such that the opening of the actuation valve **70** will increase as the actual number of revolutions of the traveling motor **51** decreases in a case where the drop in number of revolutions computed by the processor **100b** is not less than the first threshold. The changer **100c** may change the control signal based on the actual number of revolutions of the prime mover **6** in addition to the actual number of revolutions of the traveling motor **51**. More

specifically, the changer **100c** changes the control signal in a case where the target number of revolutions of the prime mover **6** operated using the number-of-revolutions operation actuator **102** is not less than a predetermined fourth threshold and, in addition, where the actual number of revolutions of the prime mover **6** is not less than a fifth threshold. The fourth threshold and the fifth threshold are preset values. The fourth threshold is defined as a value falling within a range from middle revolution to high revolution regarding the target number of revolutions of the prime mover **6**. The fifth threshold is defined as a value corresponding to, for example, the minimum value of the actual number of revolutions of the prime mover **6** required for the traveling of the working machine **1**. The fourth threshold and the fifth threshold may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(88) In the present embodiment, the changer **100c** changes the control signal based on both the actual number of revolutions of the traveling motor **51** and the target number of revolutions of the prime mover **6**; however, in a case where the changer **100c** changes the control signal based on the actual number of revolutions of the traveling motor **51** only, the changer **100c** may change the control signal regardless of the target number of revolutions of the prime mover **6** if the target number of revolutions of the prime mover **6** operated using the number-of-revolutions operation actuator **102** is not less than the predetermined fourth threshold. Changing the instruction current value by the changer **100c** will now be explained in detail.

(89) As illustrated in FIG. **1**, the working machine **1** includes a first detector **104**. The first detector **104** is, for example, a sensor configured to detect the actual number of revolutions of the traveling motor **51** (actual motor rev.). The first detector **104** is connected to the controller **100** and inputs a signal that is based on the detection (a detection signal) into the controller **100**. When the working machine **1** is equipped with a plurality of motors as the traveling motor **5**, it is possible to detect the actual number of revolutions of each of them. In the present embodiment, the traveling motor **51** includes the first traveling motor **51L** and the second traveling motor **51R**; therefore, the first detector **104** is mounted on each of the first traveling motor **51L** and the second traveling motor **51R**. The first detector (a first revolution sensor) **104** mounted on the first traveling motor **51L** detects the actual number of revolutions of the first traveling motor **51L**. The first detector (a second revolution sensor) **104** mounted on the second traveling motor **51R** detects the actual number of revolutions of the second traveling motor **51R**.

(90) Based on the detection signals inputted from the first detector **104** (the first revolution sensor and the second revolution sensor) into the controller **100**, the changer **100c** computes the actual number of revolutions of the first traveling motor **51L** and the actual number of revolutions of the second traveling motor **51R** respectively. In the present embodiment, the changer **100c** computes the actual number of revolutions of the traveling motor **51** without making a distinction between the number of revolutions at the time of normal rotation and the number of revolutions at the time of reverse rotation. For example, if the actual number of revolutions in a case of normal rotation of the traveling motor **51** is defined to be positive and the actual number of revolutions in a case of reverse rotation of the traveling motor **51** is defined to be negative, the changer **100c** computes an absolute value of the actual number of revolutions of the traveling motor **51**.

(91) The changer **100c** computes a moving average of the actual number of revolutions of the first traveling motor **51L** and a moving average of the actual number of revolutions of the second traveling motor **51R** respectively by performing moving-average calculation on the same number of pieces (n pieces) of data of the computed actual number of revolutions of the first traveling motor **51L** and the computed actual number of revolutions of the second traveling motor **51R** respectively. The changer **100c** adopts the moving average of the actual number of revolutions of the first traveling motor **51L** or the moving average of the actual number of revolutions of the second traveling motor **51R**, whichever is less, as the actual number of revolutions of the traveling motor **51**. Then, based on the moving average of the actual number of revolutions of the first traveling motor **51L** or the moving average of the actual number of revolutions of the second



traveling motor **51R**, whichever is less, the changer **100c** changes the control signal.

(92) In the description below, the moving average adopted by the changer **100c**, namely, the moving average of the actual number of revolutions of the first traveling motor **51L** or the moving average of the actual number of revolutions of the second traveling motor **51R**, whichever is less, will be simply referred to as “the actual number of revolutions of the traveling motor **51**”.

(93) In the present embodiment, it is sufficient as long as the changer **100c** changes the control signal based on the actual number of revolutions of the traveling motor **51**. In the present embodiment, for example, the changer **100c** may adopt the actual number of revolutions of the first traveling motor **51L** or the actual number of revolutions of the second traveling motor **51R**, whichever is less, as the actual number of revolutions of the traveling motor **51** without computing the moving average of the actual number of revolutions of the first traveling motor **51L** and the moving average of the actual number of revolutions of the second traveling motor **51R**.

(94) Based on a first correction coefficient (gain value) defined in relation to the actual number of revolutions of the traveling motor **51**, the changer **100c** changes the control signal expressed by the first line La. Specifically, a first function **M1** specifying a relation between the actual number of revolutions of the traveling motor **51** and the first correction coefficient is stored in the storage unit **100a**, and the changer **100c** calculates the first correction coefficient by substituting the actual number of revolutions of the traveling motor **51** detected by the first detector **104** into the first function **M1**. In the present embodiment, the changer **100c** calculates the first correction coefficient by substituting the actual number of revolutions of the traveling motor **51** adopted based on a comparison of the moving average of the actual number of revolutions of the first traveling motor **51L** and the moving average of the actual number of revolutions of the second traveling motor **51R** into the first function **M1**. FIG. 4A is a diagram illustrating an example of the first function **M1** specifying a relation between the actual number of revolutions of the traveling motor **51** and the first correction coefficient.

(95) As illustrated in FIG. 4A, the first function **M1** defines the first correction coefficient in terms of value that is not less than one, wherein a definition is given such that the first correction coefficient corresponding to a second number of revolutions, which represents the actual number of revolutions of the traveling motor **51** and is less than a first number of revolutions, is greater than the first correction coefficient corresponding to the first number of revolutions, which represents the actual number of revolutions of the traveling motor **51**. Each of the first number of revolutions and the second number of revolutions is any number of revolutions within a range specified by the first function **M1**, and the first number of revolutions is greater than the second number of revolutions. In other words, as expressed by the first function **M1**, the first correction coefficient is equal to or greater than one, decreases as the actual number of revolutions of the traveling motor **51** increases, and increases as the actual number of revolutions of the traveling motor **51** decreases. That is, since the first correction coefficient is equal to or greater than one and is never less than one, the first correction coefficient is a coefficient based on which it is possible to correct the control signal (instruction current value) expressed by the first line La to make it greater. Therefore, with the first function **M1**, the changer **100c** is capable of increasing the opening of the actuation valve **70** by performing the correction such that the instruction current value increases as the actual number of revolutions of the traveling motor **51** decreases. That is, the changer **100c** is capable of increasing the consumption horsepower of the traveling pump **50** by increasing the primary pressure as the actual number of revolutions of the traveling motor **51** decreases.

(96) As illustrated in FIG. 4A, the first function **M1** has different degrees of inclination with respect to a second threshold **Rm2** of the actual number of revolutions of the traveling motor **51**. The degree of inclination of the first function **M1** in a range where the actual number of revolutions of the traveling motor **51** is not greater than the second threshold **Rm2** is greater than the degree of inclination of the first function **M1** in a range where the actual number of revolutions of the traveling motor **51** is not less than the second threshold **Rm2**. In other words, the first function **M1**

is sloped steeper when the actual number of revolutions of the traveling motor **51** is less than the second threshold **Rm2** than when the actual number of revolutions of the traveling motor **51** is greater than the second threshold **Rm2**.

(97) This means that, on the first function **M1**, the first correction coefficient decreases with a greater amount of decrease when the actual number of revolutions of the traveling motor **51** increases from zero toward the second threshold **Rm2** than when the actual number of revolutions of the traveling motor **51** increases from the second threshold **Rm2**. To put it the other way around, on the first function **M1**, the first correction coefficient increases with a greater amount of increase when the actual number of revolutions of the traveling motor **51** decreases from the second threshold **Rm2** toward zero than when the actual number of revolutions of the traveling motor **51** decreases toward the second threshold **Rm2**.

(98) In the present embodiment, the first function **M1** is defined to have its range in which the actual number of revolutions of the traveling motor **51** is not less than a first lower limit value **Rm1** and is not greater than a first upper limit value **Rm3**, and the second threshold **Rm2** is defined as a value between the first lower limit value **Rm1** and the first upper limit value **Rm3**. That is, a part of the first function **M1** where the actual number of revolutions of the traveling motor **51** is not greater than the second threshold **Rm2** (a first part) is a partial section where the actual number of revolutions of the traveling motor **51** is not less than the first lower limit value **Rm1** and is not greater than the second threshold **Rm2** (a first section **m1**). A part of the first function **M1** where the actual number of revolutions of the traveling motor **51** is not less than the second threshold **Rm2** (a second part) is a partial section where the actual number of revolutions of the traveling motor **51** is not less than the second threshold **Rm2** and is not greater than the first upper limit value **Rm3** (a second section **m2**).

(99) The first lower limit value **Rm1** corresponds to the minimum value of the actual number of revolutions of the traveling motor **51**. In the present embodiment, the first lower limit value **Rm1** is zero. The first upper limit value **Rm3** corresponds to the maximum value of the actual number of revolutions of the traveling motor **51**. For example, the first upper limit value **Rm3** corresponds to the maximum revolution output capability of the traveling motor **51**.

(100) The second threshold **Rm2** is a preset value. The first section **m1** is defined as a section in which the actual number of revolutions of the traveling motor **51** is comparatively small, meaning a low-revolution range, as in a case where the working machine **1** is performing work such as pushing dirt using its bucket. In the present embodiment, the first section **m1** corresponds to a speed range for a case where the traveling motor **51** is operating at the first speed (low-speed range), and the second threshold **Rm2** is a value corresponding to any actual number of revolutions in a speed range for a case where the traveling motor **51** is operating at the first speed. The second threshold **Rm2** may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(101) In both of the first section **m1** and the second section **m2**, the first correction coefficient is in proportion to the actual number of revolutions of the traveling motor **51**. As illustrated in FIG. 4A, on the first function **M1**, the first correction coefficient is 1.6 (160%) when the actual number of revolutions of the traveling motor **51** is at the first lower limit value **Rm1**. On the first function **M1**, the first correction coefficient is 1.2 (120%) when the actual number of revolutions of the traveling motor **51** is at the second threshold **Rm2**. On the first function **M1**, the first correction coefficient is 1.0 (100%) when the actual number of revolutions of the traveling motor **51** is at the first upper limit value **Rm3**.

(102) That is, in the first section **m1** of the first function **M1** illustrated in FIG. 4A, the first correction coefficient decreases substantially linearly from 1.6 (160%) to 1.2 (120%) as the actual number of revolutions of the traveling motor **51** increases, and increases substantially linearly from 1.2 (120%) to 1.6 (160%) as the actual number of revolutions of the traveling motor **51** decreases.

(103) On the other hand, in the second section **m2** of the first function **M1** illustrated in FIG. 4A,

the first correction coefficient decreases substantially linearly from 1.2 (120%) to 1.0 (100%) as the actual number of revolutions of the traveling motor **51** increases, and increases substantially linearly from 1.0 (100%) to 1.2 (120%) as the actual number of revolutions of the traveling motor **51** decreases.

(104) Therefore, the changer **100c** is capable of increasing the delivery flow rate of the traveling pump **50** (consumption horsepower of the traveling pump **50**) by increasing the output of the traveling pump **50** when the actual number of revolutions of the traveling motor **51** is comparatively small as in a case where of performing work such as pushing dirt. That is, by increasing the consumption horsepower of the traveling pump **50**, it is possible to reduce the actual number of revolutions of the prime mover **6** that is in balance with the delivery flow rate of the traveling pump **50**. By this means, it is possible to prevent the occurrence of a phenomenon that, in work that involves opening the traveling relief valve **71**, for example, when pushing dirt, the consumption horsepower of the traveling pump **50** decreases due to the influence of the swash-plate characteristics of the traveling pump **50**, resulting in a large actual number of revolutions of the prime mover **6** that is in balance with the delivery flow rate of the traveling pump **50**. Consequently, it is possible to give, to the operator, an operational feeling that the working machine **1** is performing work with sufficient performance when pushing dirt or the like, and it is possible to continue the work by increasing the delivery flow rate of the traveling pump **50** even when there is an increase in traveling load.

(105) Though some values have been described in the above explanation of the first correction coefficient, they are mere illustrative values mentioned just for the purpose of giving an example. The first correction coefficient may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(106) In the present embodiment, a case where the first correction coefficient is in proportion to the actual number of revolutions of the traveling motor **51** both in the first section **m1** and in the second section **m2** has been taken as an example; however, it is sufficient as long as the first function **M1** has different degrees of inclination with respect to a second threshold **Rm2** of the actual number of revolutions of the traveling motor **51**. That is, the first function **M1** may be a substantially “drawing a curve” function.

(107) Based on a second correction coefficient (gain value) defined in relation to the actual number of revolutions of the prime mover **6** in addition to the first correction coefficient, the changer **100c** changes the control signal expressed by the first line **La**. Specifically, a second function **M2** specifying a relation between the actual number of revolutions of the prime mover **6** and the second correction coefficient is stored in the storage unit **100a**, and the changer **100c** calculates the second correction coefficient by substituting the actual number of revolutions of the prime mover **6** detected by the second detector **103** into the second function **M2**. FIG. 4B is a diagram illustrating an example of the second function **M2** specifying a relation between the actual number of revolutions of the prime mover **6** and the second correction coefficient.

(108) As illustrated in FIG. 4B, the second function **M2** defines the second correction coefficient as a value equal to one when the actual number of revolutions of the prime mover **6** is not less than a third threshold **Re2**, and defines the second correction coefficient in terms of value that is less than one when the actual number of revolutions of the prime mover **6** is less than the third threshold **Re2**. Therefore, according to the second function **M2**, at least, the changer **100c** never changes the instruction current value such that the opening of the actuation valve **70** will increase, and is capable of decreasing the opening of the actuation valve **70** when the actual number of revolutions of the prime mover **6** is less than the third threshold **Re2** by changing the instruction current value in such a way as to decrease as the actual number of revolutions of the prime mover **6** decreases. That is, when the actual number of revolutions of the prime mover **6** is less than the third threshold **Re2**, the changer **100c** is capable of decreasing the consumption horsepower of the traveling pump **50** by reducing the primary pressure as the actual number of revolutions of the prime mover **6**

decreases.

(109) In the present embodiment, the second function M2 is defined to have its range in which the actual number of revolutions of the prime mover **6** is not less than a second lower limit value Re1 and is not greater than a second upper limit value Re3, and the third threshold Re2 is defined as a value between the second lower limit value Re1 and the second upper limit value Re3. That is, a part of the second function M2 where the actual number of revolutions of the prime mover **6** is less than the third threshold Re2 (a third part) is a partial section where the actual number of revolutions of the prime mover **6** is not less than the second lower limit value Re1 and is less than the third threshold Re2 (a third section m3). A part of the second function M2 where the actual number of revolutions of the prime mover **6** is not less than the third threshold Re2 (a fourth part) is a partial section where the actual number of revolutions of the prime mover **6** is not less than the third threshold Re2 and is not greater than the second upper limit value Re3 (a fourth section m4).

(110) The second lower limit value Re1 is defined as a value equal to the fifth threshold. That is, in the present embodiment, the second lower limit value Re1 corresponds to, for example, the minimum value of the actual number of revolutions of the prime mover **6** required for the traveling of the working machine **1**. The second upper limit value Re3 corresponds to the maximum value of the actual number of revolutions of the prime mover **6**. For example, the second upper limit value Re3 corresponds to the maximum revolution output capability of the prime mover **6**. The third threshold Re2 is a preset value. The third section m3 is defined as a section in which the actual number of revolutions of the prime mover **6** does not go beyond a middle-revolution range as in a case where the working machine **1** is climbing an ascending slope. In the present embodiment, the third threshold Re2 is defined as a value that is less than the fourth threshold. The third threshold Re2 may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(111) In the third section m3, in which the actual number of revolutions of the prime mover **6** is less than the third threshold Re2, the value of the second correction coefficient is less than one, and the second correction coefficient is in proportion to the actual number of revolutions of the prime mover **6**. More specifically, in the third section m3, the second correction coefficient increases as the actual number of revolutions of the prime mover **6** increases, and decreases as the actual number of revolutions of the prime mover **6** decreases. For example, the second correction coefficient is defined to take a value within a range from 0.9 (90%) to 1.0 (100%) in the third section m3 of the second function M2.

(112) In the fourth section m4, in which the actual number of revolutions of the prime mover **6** is not less than the third threshold Re2, the value of the second correction coefficient is one. That is, in the fourth section m4, in which the actual number of revolutions of the prime mover **6** is not less than the third threshold Re2, the changer **100c** does not change the control signal, based on the second correction coefficient.

(113) That is, in the third section m3 of the second function M2 illustrated in FIG. 4B, the second correction coefficient increases substantially linearly from 0.9 (90%) to 1.0 (100%) as the actual number of revolutions of the prime mover **6** increases, and decreases substantially linearly from 1.0 (100%) to 0.9 (90%) as the actual number of revolutions of the prime mover **6** decreases.

(114) On the other hand, in the fourth section m4 of the second function M2 illustrated in FIG. 4B, the second correction coefficient is constant at 1.0 (100%).

(115) Therefore, when the number of revolutions of the prime mover **6** is less than the third threshold Re2 and is comparatively small, the changer **100c** is capable of changing the first line La in a direction in which the primary pressure outputted from the actuation valve **70** decreases. By this means, it is possible to reduce the output of the traveling pump **50** and thus prevent the engine from stalling.

(116) Though some values have been described in the above explanation of the second correction coefficient, they are mere illustrative values mentioned just for the purpose of giving an example.

The second correction coefficient may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(117) The changer **100c** changes the control signal expressed by the first line La by means of a third correction coefficient that is based on the first correction coefficient and the second correction coefficient. Specifically, the changer **100c** computes the third correction coefficient based on a product of the first correction coefficient and the second correction coefficient. In a case where the value of the third correction coefficient computed by the changer **100c** is greater than one (100%), when the changer **100c** multiplies the instruction current value by the third correction coefficient, the instruction current value after the change will be greater than the instruction current value before the change. Therefore, in a case where the value of the third correction coefficient acquired by the changer **100c** is greater than one, the change is made in a direction in which the primary pressure outputted from the actuation valve **70** to which the instruction current value after the change is outputted will become greater than before the change. In a case where the value of the third correction coefficient computed by the changer **100c** is equal to one (100%), even when the changer **100c** multiplies the instruction current value by the third correction coefficient, the instruction current value after the change will remain the same.

(118) The changer **100c** changes the control signal expressed by the first line La in a case where the value of the third correction coefficient is greater than one and does not change the control signal expressed by the first line La in a case where the value of the third correction coefficient is not greater than one. By this means, it is possible to prevent the changer **100c** from changing the first line La excessively in a direction in which the primary pressure outputted from the actuation valve **70** decreases. That is, it is possible to prevent the traveling power of the traveling motor **51** from being reduced excessively due to changing the first line La by the changer **100c** when the actual number of revolutions of the prime mover **6** is less than the third threshold Re2.

(119) As described above, in the present embodiment, the changer **100c** does not change the control signal expressed by the first line La in a case where the value of the third correction coefficient is not greater than one; however, if a higher priority should be given to engine stall suppression, the changer **100c** may change the control signal expressed by the first line La with multiplication of the instruction current value by the third correction coefficient in a case where the value of the third correction coefficient is not greater than one.

(120) In this “stall-suppression-first” case, when the changer **100c** multiplies the instruction current value by the third correction coefficient, with the value of the third correction coefficient acquired by the changer **100c** being less than one (100%), the instruction current value after the change will be less than the instruction current value before the change. Therefore, in a case where the value of the third correction coefficient acquired by the changer **100c** is less than one, the change is made in a direction in which the primary pressure outputted from the actuation valve **70** to which the instruction current value after the change is outputted will become less than before the change.

(121) FIG. 5A is an operation flowchart illustrating the flow of operation of the controller **100** for changing the control signal (instruction current value). With reference to FIG. 5A, a series of steps for changing the control signal by the controller **100** will now be explained.

(122) First, based on the target number of revolutions of the prime mover **6** operated using the number-of-revolutions operation actuator **102** and the actual number of revolutions of the prime mover **6** detected by the second detector **103**, the processor **100b** computes the drop in number of revolutions (S1). Specifically, the processor **100b** acquires information on the target number of revolutions of the prime mover **6** operated using the number-of-revolutions operation actuator **102** and the actual number of revolutions of the prime mover **6** detected by the second detector **103**, and computes the drop in number of revolutions by subtracting the actual number of revolutions of the prime mover **6** from the target number of revolutions of the prime mover **6**.

(123) After computing the drop in number of revolutions (S1), the processor **100b** checks whether the drop in number of revolutions is less than the first threshold or not (S2). If the drop in number

of revolutions is confirmed to be less than the first threshold (S2: Yes), the processor **100b** acquires the second line Lb from the storage unit **100a** (S3). After acquiring the second line Lb from the storage unit **100a** (S3), based on the acquired second line Lb, the processor **100b** acquires the instruction current value corresponding to the actual number of revolutions detected by the second detector **103** (S4).

(124) After acquiring the instruction current value corresponding to the actual number of revolutions (S4), the processor **100b** defines the acquired instruction current value as the control signal outputted by the controller **100** to the actuation valve **70** (S5).

(125) On the other hand, if the drop in number of revolutions is confirmed to be not less than the first threshold, namely, if the drop in number of revolutions is confirmed to be equal to or greater than the first threshold (S2: No), the processor **100b** acquires the first line La from the storage unit **100a** (S6). After acquiring the first line La from the storage unit **100a** (S6), based on the acquired first line La, the processor **100b** acquires the instruction current value corresponding to the actual number of revolutions detected by the second detector **103** (S7).

(126) After the acquisition of the instruction current value corresponding to the actual number of revolutions by the processor **100b** (S7), the changer **100c** acquires the first function M1 from the storage unit **100a** (S8). After acquiring the first function M1 from the storage unit **100a** (S8), based on a detection signal of the first detector **104**, the changer **100c** computes the actual number of revolutions of the traveling motor **51** (S9). After computing the actual number of revolutions of the traveling motor **51** (S9), the changer **100c** calculates the first correction coefficient by substituting the actual number of revolutions of the traveling motor **51** into the first function M1 (S10).

(127) After calculating the first correction coefficient (S10), the changer **100c** acquires the second function M2 from the storage unit **100a** (S11). After acquiring the second function M2 from the storage unit **100a** (S11), the changer **100c** calculates the second correction coefficient by substituting the actual number of revolutions of the prime mover **6** into the second function M2 (S12).

(128) After calculating the second correction coefficient (S12), the changer **100c** computes the third correction coefficient based on a product of the first correction coefficient and the second correction coefficient (S13). After computing the third correction coefficient (S13), the changer **100c** checks whether the value of the computed third correction coefficient is greater than one or not (S14).

(129) If the value of the third correction coefficient is confirmed to be greater than one (S14: Yes), the changer **100c** changes the control signal expressed by the first line La by multiplying the instruction current value having been acquired by the processor **100b** by the third correction coefficient (S15). After the changing of the control signal by the changer **100c** (S15), the processor **100b** defines the instruction current value after the change (control signal after the change) as the control signal outputted by the controller **100** to the actuation valve **70** (S16).

(130) On the other hand, if the value of the third correction coefficient is confirmed to be not greater than one, namely, if the value of the third correction coefficient is confirmed to be equal to or less than one (S14: No), the changer **100c** does not change the control signal expressed by the first line La by means of the third correction coefficient (S17). That is, the processor **100b** defines the instruction current value (control signal) expressed by the first line La as the control signal outputted by the controller **100** to the actuation valve **70** (S18).

(131) In a modification example in which the changer **100c** changes the control signal expressed by the first line La in a case where the value of the third correction coefficient is less than one, the controller **100** operates as illustrated in FIG. 5B for correcting the control signal. FIG. 5B is an operation flowchart illustrating the flow of operation of the controller **100** for changing the control signal according to a first modification example. Specifically, as illustrated in FIG. 5B, the flow of operation of the controller **100** according to the modification example is different from the operation flow illustrated in FIG. 5A in that S14 to S18 are replaced with S19 to S23. More

specifically, in the modification example illustrated in FIG. 5B, after computing the third correction coefficient (S13), the changer **100c** checks whether the value of the computed third correction coefficient is equal to one or not (S19).

(132) If the value of the third correction coefficient is confirmed to be equal to one (S19: Yes), the changer **100c** does not change the control signal expressed by the first line La by means of the third correction coefficient (S20). That is, the processor **100b** defines the instruction current value expressed by the first line La as the control signal outputted by the controller **100** to the actuation valve **70** (S21).

(133) On the other hand, if the value of the third correction coefficient is confirmed to be not equal to one (S19: No), the changer **100c** changes the control signal expressed by the first line La by multiplying the instruction current value by the third correction coefficient (S22). After the changing of the control signal by the changer **100c** (S22), the processor **100b** defines the instruction current value after the change as the control signal outputted by the controller **100** to the actuation valve **70** (S23).

(134) The controller **100** may be configured to be switchable between a plurality of modes, and the storage unit **100a** may store a plurality of first functions M1 and a plurality of second functions M2 corresponding to the plurality of modes. FIG. 4C is a diagram illustrating an example of respective first functions M1 for a plurality of modes in a modification example in which the controller **100** is configured to be switchable between the plurality of modes. FIG. 4D is a diagram illustrating an example of respective second functions M2 for the plurality of modes in this modification example. As illustrated in FIG. 4C, the first functions M1 are different from one another in terms of inclination at least partially. As illustrated in FIG. 4C, the second functions M2 are different from one another in terms of inclination at least partially. Based on the first function M1 that corresponds to the mode of the controller **100**, the changer **100c** calculates the first correction coefficient. In addition, based on the second function M2 that corresponds to the mode of the controller **100**, the changer **100c** calculates the second correction coefficient.

(135) That is, configuring the controller **100** to be switchable between plural modes enables the changer **100c** to change a control signal in a different manner according to the mode. In the description below, a case where both of first and second functions M1 and M2 correspond to a plurality of modes and where a plurality of first functions M1 and a plurality of second functions M2 are stored in the storage unit **100a** will be taken as an example. The example may be modified such that either first functions M1 only or second functions M2 only correspond to a plurality of modes. In this case, the storage unit **100a** may store a plurality of first functions M1 corresponding to a plurality of modes and a single second function M2 or store a single first function M1 and a plurality of second functions M2 corresponding to a plurality of modes. The controller **100** may be configured such that mode switching can be performed separately for a plurality of modes corresponding to a plurality of first functions M1 and a plurality of modes corresponding to a plurality of second functions M2. The number of first functions M1 and the number of second functions M2 do not necessarily have to match each other. The combination of them is also not limited.

(136) As illustrated in FIG. 1, the working machine **1** includes a switcher (operation actuator) **105**. The switcher **105** is a selector switch operable for switching the mode of the controller **100**. The switcher **105** is connected to the controller **100**, is operated by the operator or the like, and inputs an operation signal into the controller **100**. In the present embodiment, the switcher **105** is a display image displayed on a display **110**. The display **110** is a device connected to the controller **100** such that communication can be performed therebetween and configured to display various kinds of information regarding the working machine **1** to assist the traveling and work of the working machine **1**. The display **110** is, for example, a traveling assistance device provided near the operator's seat **8**. The display **110** is further connected to devices of the working machine **1** such that wired or wireless communication can be performed therebetween, and is therefore capable of

transmitting information thereto and receiving information therefrom bi-directionally.

(137) FIG. 6 is a diagram illustrating an example of the display **110** according to the modification example, and a switching screen D1 displayed on its display unit **111**. As illustrated in FIG. 6, the display **110** includes the display unit **111**. The display unit **111** is any one of a liquid crystal panel, a touch panel, and other kinds of panel. Various kinds of information for assisting the traveling and work of the working machine **1** can be displayed thereon.

(138) In response to predetermined operation performed by the operator, the display unit **111** displays the switching screen D1. The switching screen D1 displays the switcher **105**. The switcher **105** includes a plurality of selection buttons **105a** to accept operation by the operator. In the present embodiment, the operator performs selection operation by touching one of the plurality of selection buttons **105a** displayed in the switching screen D1. When one of the selection buttons **105a** is selected by the operator, the display **110** outputs operation information that is based on the selection operation to the controller **100**. Though the switcher **105** according to the present embodiment includes the plurality of selection buttons **105a** displayed on the display unit **111**, the switcher **105** may be embodied in any other form as long as it is connected to the controller **100** and enables mode switching operation. For example, the switcher **105** may be a dial connected to the controller **100** and having a plurality of switching positions, a plurality of buttons, etc.

(139) The display unit **111** is capable of displaying which one of the modes the controller **100** is in now. FIG. 7 is a diagram for explaining a mode display portion **111a** displayed by the display **110** and the display unit **111** according to the modification example. For example, as illustrated in FIG. 7, there is a mode display portion **111a** at a top area dl of a screen D2 displayed by the display unit **111**. The mode display portion **111a** shows the current mode of the controller **100** by means of a character string. In the example illustrated in FIG. 7, the current mode of the controller **100** is shown by means of a character string. However, the mode display is not limited to character-string display. The display unit **111** may display the current mode of the controller **100** by means of graphics such as any kind of icon or the like.

(140) In the present embodiment, the controller **100** has a first mode, a second mode, and a third mode as the plurality of modes and is configured to be switchable between them. The first mode is, for example, a mode that is designed to offer enhanced operational feeling of the working machine **1** for a case where the actual number of revolutions of the traveling motor **51** is comparatively small. That is, the first mode is a mode for reducing the number of revolutions of the engine that is in balance with the delivery flow rate of the traveling pump **50** more by boosting the consumption horsepower of the traveling pump **50** when the actual number of revolutions of the traveling motor **51** is comparatively small. The third mode is a mode for giving a higher priority to engine stall suppression than to operational feeling of the working machine **1**. The second mode is an intermediate mode between the first mode and the third mode. The first function M1 for each of the plurality of modes (first, second, and third modes) will now be explained.

(141) FIG. 4C is a diagram illustrating an example of respective first functions M1 for the plurality of modes in the modification example. As illustrated in FIG. 4C, the first functions M1 have, as partial inclination of them different from mode to mode, respective slopes different from one another at the first section m1. In FIG. 4C, a first section m1a of the first function M1 in the first mode is indicated by a solid line, a first section m1b of the first function M1 in the second mode is indicated by an alternate-long-and-single-short-dashed line, and a first section m1c of the first function M1 in the third mode is indicated by an alternate-long-and-double-short-dashed line. The first functions M1 have the same value of the first correction coefficient when the actual number of revolutions of the traveling motor **51** is at the second threshold Rm2.

(142) As illustrated in FIG. 4C, the slope at the first section m1 in the first mode is the steepest of those in the three modes, and the slope at the first section m1 in the third mode is the gentlest of those in the three modes. That is, an amount of change in the first correction coefficient in the first mode relative to the actual number of revolutions of the traveling motor **51** is the largest among the



three modes, and the first correction coefficient in the first mode works to correct the instruction current value in such a way as to make the primary pressure at the actuation valve **70** greater than in the other modes. An amount of change in the first correction coefficient in the third mode relative to the actual number of revolutions of the traveling motor **51** is the smallest among the three modes, and the first correction coefficient in the third mode works to correct the instruction current value in such a way as to make the primary pressure at the actuation valve **70** less than in the other modes. That is, in the present embodiment, given the same actual number of revolutions of the traveling motor **51**, the first correction coefficient is greater in the order of the third mode, the second mode, and the first mode.

(143) In an example of the first function **M1** corresponding to the first mode illustrated in FIG. **4C**, the first correction coefficient is 1.6 (160%) when the actual number of revolutions of the traveling motor **51** is at the first lower limit value **Rm1**. Therefore, in the first section **m1a**, the first correction coefficient in the first mode decreases substantially linearly from 1.6 (160%) to 1.2 (120%) as the actual number of revolutions of the traveling motor **51** increases, and increases substantially linearly from 1.2 (120%) to 1.6 (160%) as the actual number of revolutions of the traveling motor **51** decreases.

(144) In an example of the first function **M1** corresponding to the second mode illustrated in FIG. **4C**, the first correction coefficient is 1.5 (150%) when the actual number of revolutions of the traveling motor **51** is at the first lower limit value **Rm1**. Therefore, in the first section **m1b**, the first correction coefficient in the second mode decreases substantially linearly from 1.5 (150%) to 1.2 (120%) as the actual number of revolutions of the traveling motor **51** increases, and increases substantially linearly from 1.2 (120%) to 1.5 (150%) as the actual number of revolutions of the traveling motor **51** decreases.

(145) In an example of the first function **M1** corresponding to the third mode illustrated in FIG. **4C**, the first correction coefficient is 1.4 (140%) when the actual number of revolutions of the traveling motor **51** is at the first lower limit value **Rm1**. Therefore, in the first section **m1c**, the first correction coefficient in the third mode decreases substantially linearly from 1.4 (140%) to 1.2 (120%) as the actual number of revolutions of the traveling motor **51** increases, and increases substantially linearly from 1.2 (120%) to 1.4 (140%) as the actual number of revolutions of the traveling motor **51** decreases.

(146) Though some values of the first correction coefficient have been described in the above explanation of the plurality of modes, they are mere illustrative values mentioned just for the purpose of giving an example. It is sufficient as long as, at least, the first correction coefficient is greater in the order of the third mode, the second mode, and the first mode. The first correction coefficient in the plurality of modes may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(147) FIG. **4D** is a diagram illustrating an example of respective second functions **M2** for the plurality of modes in the modification example. As illustrated in FIG. **4D**, the second functions **M2** have, as partial inclination of them different from mode to mode, respective slopes different from one another at the third section **m3**. In FIG. **4D**, a third section **m3a** of the second function **M2** in the first mode is indicated by a solid line, a third section **m3b** of the second function **M2** in the second mode is indicated by an alternate-long-and-single-short-dashed line, and a third section **m3c** of the second function **M2** in the third mode is indicated by an alternate-long-and-double-short-dashed line. The second functions **M2** have the same value of the second correction coefficient when the actual number of revolutions of the prime mover **6** is at the third threshold **Re2**.

(148) As illustrated in FIG. **4D**, the slope at the third section **m3** in the first mode is the gentlest of those in the three modes, and the slope at the third section **m3** in the third mode is the steepest of those in the three modes. That is, an amount of change in the second correction coefficient in the first mode relative to the actual number of revolutions of the prime mover **6** is the smallest among the three modes, and the second correction coefficient in the first mode works to correct the

instruction current value in such a way as to make the primary pressure at the actuation valve **70** greater than in the other modes. An amount of change in the second correction coefficient in the third mode relative to the actual number of revolutions of the prime mover **6** is the largest among the three modes, and the second correction coefficient in the third mode works to correct the instruction current value in such a way as to make the primary pressure at the actuation valve **70** less than in the other modes. That is, in the present embodiment, given the same actual number of revolutions of the prime mover **6**, the second correction coefficient is less in the order of the first mode, the second mode, and the third mode.

(149) In an example of the second function **M2** corresponding to the first mode illustrated in FIG. **4D**, the second correction coefficient is 0.9 (90%) when the actual number of revolutions of the prime mover **6** is at the second lower limit value **Re1**. Therefore, in the third section **m3a**, the second correction coefficient in the first mode increases substantially linearly from 0.9 (90%) to 1.0 (100%) as the actual number of revolutions of the prime mover **6** increases, and decreases substantially linearly from 1.0 (100%) to 0.9 (90%) as the actual number of revolutions of the prime mover **6** decreases.

(150) In an example of the second function **M2** corresponding to the second mode illustrated in FIG. **4D**, the second correction coefficient is 0.8 (80%) when the actual number of revolutions of the prime mover **6** is at the second lower limit value **Re1**. Therefore, in the third section **m3b**, the second correction coefficient in the second mode increases substantially linearly from 0.8 (80%) to 1.0 (100%) as the actual number of revolutions of the prime mover **6** increases, and decreases substantially linearly from 1.0 (100%) to 0.8 (80%) as the actual number of revolutions of the prime mover **6** decreases.

(151) In an example of the second function **M2** corresponding to the third mode illustrated in FIG. **4D**, the second correction coefficient is 0.5 (50%) when the actual number of revolutions of the prime mover **6** is at the second lower limit value **Re1**. Therefore, in the third section **m3c**, the second correction coefficient in the third mode increases substantially linearly from 0.5 (50%) to 1.0 (100%) as the actual number of revolutions of the prime mover **6** increases, and decreases substantially linearly from 1.0 (100%) to 0.5 (50%) as the actual number of revolutions of the prime mover **6** decreases.

(152) Though some values of the second correction coefficient have been described in the above explanation of the plurality of modes, they are mere illustrative values mentioned just for the purpose of giving an example. It is sufficient as long as, at least, the second correction coefficient is less in the order of the first mode, the second mode, and the third mode. The second correction coefficient in the plurality of modes may be configured to be changeable by operating an operation switch connected to the controller **100**, operating a terminal, or the like.

(153) FIG. **5C** is an operation flowchart illustrating the flow of operation of the controller **100** for changing the control signal according to a second modification example. Described below with reference to FIG. **5C** is a series of steps for changing the control signal (instruction current value) when the controller **100** is configured to be switchable between the plurality of modes. In this case, the flow of operation of the controller **100** for correcting the control signal is different from the operation flow illustrated in FIG. **5A** in that, as illustrated in FIG. **5C**, **S8** is replaced with **S30** and **S31**, and **S11** is replaced with **S32**. The description below will be given with a focus on **S30-S31** and **S32**, and the other steps will not be explained.

(154) After the acquisition of the instruction current value corresponding to the actual number of revolutions by the processor **100b** (**S7**), the changer **100c** checks the current mode of the controller **100** (**S30**). Specifically, based on operation information of the switcher **105**, the changer **100c** checks which one of the modes the controller **100** is in now. After checking the current mode of the controller **100** (**S30**), the changer **100c** acquires the first function **M1** corresponding to the current mode from the storage unit **100a** (**S31**).

(155) After acquiring the first function **M1** from the storage unit **100a** (**S31**), based on a detection

signal of the first detector **104**, the changer **100c** computes the actual number of revolutions of the traveling motor **51** (S9).

(156) After calculating the first correction coefficient (S10), the changer **100c** acquires the second function M2 corresponding to the current mode from the storage unit **100a** (S32).

(157) After acquiring the second function M2 from the storage unit **100a** (S32), the changer **100c** calculates the second correction coefficient by substituting the actual number of revolutions of the prime mover **6** into the second function M2 (S12).

(158) In the foregoing embodiment, the actuation valve **70** is provided upstream of the operation valves **56** (delivery fluid passage **40**). However, for example, the actuation valve **70** may be provided somewhere on the fifth traveling fluid passage **42e** instead.

(159) Alternatively, as illustrated in FIG. **8**, the actuation valve **70** may be provided on the traveling fluid passage **42** connected to the traveling pump **50** (the first traveling pump SOL and the second traveling pump **50R**). Specifically, fluid passages **44** branching off from the first traveling fluid passage **42a**, the second traveling fluid passage **42b**, the third traveling fluid passage **42c**, and the fourth traveling fluid passage **42d** respectively may be provided, and the actuation valve **70** such as a variable relief valve or a proportional solenoid valve may be provided on the fluid passages **44**, and the opening of the actuation valve **70** may be controlled by means of a first control signal and a second control signal.

(160) In the foregoing embodiment, the manipulator **54** is a hydraulic-type device configured to change pilot pressure acting on the traveling pump **50** (the first traveling pump SOL and the second traveling pump **50R**) by means of the operation valves **56**. However, as illustrated in FIG. **9**, the manipulator **54** may be an electric-type device configured to operate electrically.

(161) As illustrated in FIG. **9**, the manipulator **54** includes the operation member **55** configured to be swung in the left-right direction (machine-body width direction) or the front-rear direction selectively and the operation valves **56** (the first pilot valve **56a**, the second pilot valve **56b**, the third pilot valve **56c**, and the fourth pilot valve **56d**) that are proportional solenoid valves. An operation detection sensor configured to detect an operation amount and an operation direction of the operation member **55** is connected to the controller **100**. Based on the operation amount and the operation direction detected by the operation detection sensor, the controller **100** controls the operation valves **56** (the first pilot valve **56a**, the second pilot valve **56b**, the third pilot valve **56c**, and the fourth pilot valve **56d**).

(162) When the operation member **55** is operated forward (in the direction indicated by the arrow **A1**; see FIG. **1**), the controller **100** outputs control signals to the first pilot valve **56a** and the third pilot valve **56c** to cause the swash plates of the first traveling pump SOL and the second traveling pump SOR to tilt in the normal (forward) direction.

(163) When the operation member **55** is operated rearward (in the direction indicated by the arrow **A2**; see FIG. **1**), the controller **100** outputs control signals to the second pilot valve **56b** and the fourth pilot valve **56d** to cause the swash plates of the first traveling pump SOL and the second traveling pump SOR to tilt in the reverse (rearward) direction.

(164) When the operation member **55** is operated leftward (in the direction indicated by the arrow **A3**; see FIG. **1**), the controller **100** outputs control signals to the second pilot valve **56b** and the third pilot valve **56c** to cause the swash plate of the first traveling pump SOL to tilt in the reverse direction and cause the swash plate of the second traveling pump SOR to tilt in the normal direction.

(165) When the operation member **55** is operated rightward (in the direction indicated by the arrow **A4**; see FIG. **1**), the controller **100** outputs control signals to the first pilot valve **56a** and the fourth pilot valve **56d** to cause the swash plate of the first traveling pump SOL to tilt in the normal direction and cause the swash plate of the second traveling pump SOR to tilt in the reverse direction.

(166) The working machine **1** described above includes: the prime mover **6**; the traveling pump **50**

configured to operate by power of the prime mover **6** and deliver a hydraulic fluid; the traveling motor **51** capable of rotating using the hydraulic fluid delivered by the traveling pump **50**; the operation valve **56** capable of changing pilot pressure of a pilot fluid outputted to the traveling pump **50** in response to operation of the operation member **55**; the actuation valve **70** configured to operate based on a control signal and capable of changing primary pressure that is the pilot pressure of the pilot fluid supplied to the operation valve **56**; the controller **100** configured or programmed to control the opening of the actuation valve **70** by outputting the control signal to the actuation valve **70**; and the first detector **104** configured to detect the actual number of revolutions of the traveling motor **51**, wherein the controller **100** includes the changer **100c** configured or programmed to change setting of the control signal such that the opening of the actuation valve **70** increases as the actual number of revolutions of the traveling motor **51** decreases. With this configuration, based on the actual number of revolutions of the traveling motor **51**, the controller **100** is able to change the setting of the control signal outputted to the actuation valve **70** by means of the changer **100c**. Therefore, based on the actual number of revolutions of the traveling motor **51**, the controller **100** is able to change the primary pressure that is the pilot pressure of the pilot fluid supplied by the actuation valve **70** to the operation valve **56**. By this means, it is possible to give an operational feeling that the working machine **1** is performing work with sufficient performance to the operator according to the actual number of revolutions of the traveling motor **51**.

(167) The working machine **1** further includes: the number-of-revolutions operation actuator **102** configured for the operator to operate a target number of revolutions of the prime mover **6**; the second detector **103** configured to detect an actual number of revolutions of the prime mover **6**; and the storage unit **100a** in which the first line La and the second line Lb are stored, the first line La being a line defining the control signal based on the actual number of revolutions of the prime mover **6** in a case where a difference between the target number of revolutions of the prime mover **6** and the actual number of revolutions of the prime mover **6** is not less than a first threshold, the second line Lb being a line defining the control signal to be greater than defined by the first line La in a case where the difference is less than the first threshold, wherein the changer **100c** changes the first line La by changing the control signal expressed by the first line La such that the opening of the actuation valve **70** increases as the actual number of revolutions of the traveling motor **51** decreases. With this configuration, depending on which of two cases applies, one of the two cases being a case where the prime mover **6** is under light load (the difference between the target number and the actual number is less than the first threshold), and the other being a case where the prime mover **6** is under heavy load (the difference between the target number and the actual number is not less than the first threshold), the changer **100c** is able to change the setting of the control signal outputted to the actuation valve **70** appropriately based on the actual number of revolutions of the traveling motor **51**.

(168) Based on a first correction coefficient defined in relation to the actual number of revolutions of the traveling motor **51**, the changer **100c** changes the control signal expressed by the first line La. With this configuration, the changer **100c** is able to change the setting of the control signal outputted to the actuation valve **70** by using the first correction coefficient defined in relation to the actual number of revolutions of the traveling motor **51**. Therefore, it is possible to change the control signal easily and appropriately.

(169) The first function M1 specifying a relation between the actual number of revolutions of the traveling motor **51** and the first correction coefficient is stored in the storage unit **100a**, and the changer **100c** calculates the first correction coefficient by substituting the actual number of revolutions of the traveling motor **51** detected by the first detector **104** into the first function M1. With this configuration, the changer **100c** is able to calculate the correction coefficient easily and accurately by using the function specifying the relation between the actual number of revolutions of the traveling motor **51** and the first correction coefficient.

(170) The first function **M1** defines the first correction coefficient in terms of value that is not less than one, wherein a definition is given such that the first correction coefficient corresponding to a second number of revolutions, which represents the actual number of revolutions of the traveling motor **51** and is less than a first number of revolutions, is greater than the first correction coefficient corresponding to the first number of revolutions, which represents the actual number of revolutions of the traveling motor **51**. With this configuration, since the first correction coefficient is defined in terms of value that is not less than one, the changer **100c** is able to increase the consumption horsepower of the traveling pump **50** by changing the first line **La** in a direction in which the primary pressure outputted from the actuation valve **70** increases. More particularly, the changer **100c** is able to increase the consumption horsepower of the traveling pump **50** to reduce the number of revolutions of the engine that is in balance with the delivery flow rate of the traveling pump **50** when the actual number of revolutions of the traveling motor **51** is comparatively small. By this means, the controller **100** is able to prevent a decrease in traveling power in a high-revolution range while preventing a phenomenon in which the prime mover **6** remains at high revolution, and give an operational feeling that the working machine **1** is performing work with sufficient performance to the operator. Moreover, even when there is an increase in traveling load, it is possible to continue the work by increasing the delivery flow rate of the traveling pump **50**.

(171) The first function **M1** has different degrees of inclination with respect to a second threshold **Rm2** of the actual number of revolutions of the traveling motor **51**. The degree of inclination of the first function **M1** in a range where the actual number of revolutions of the traveling motor **51** is not greater than the second threshold **Rm2** is greater than the degree of inclination of the first function **M1** in a range where the actual number of revolutions of the traveling motor **51** is not less than the second threshold **Rm2**. With this configuration, the changer **100c** is able to increase the consumption horsepower of the traveling pump **50** by increasing the output of the traveling pump **50** when the actual number of revolutions of the traveling motor **51** is comparatively small as in a case where of performing work such as pushing dirt. Therefore, it is possible to reduce the number of revolutions of the engine that is in balance therewith, and give an operational feeling that the working machine **1** is performing work with sufficient performance to the operator.

(172) The controller **100** is switchable between a plurality of modes, and a plurality of first functions **M1** different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit **100a**, and the changer **100c** calculates the first correction coefficient based on, among the plurality of first functions **M1**, a first function **M1** that corresponds to a mode of the controller **100**. With this configuration, the controller is able to change the mode depending on the degree of priority, for example, “engine stall suppression first”, “operational feeling first”, “traveling power first”, and the like.

(173) Based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover **6** in addition to the first correction coefficient, the changer **100c** changes the control signal expressed by the first line **La**. With this configuration, the changer **100c** is able to change the setting of the control signal outputted to the actuation valve **70** by using the second correction coefficient defined in relation to the actual number of revolutions of the prime mover **6**. Therefore, it is possible to change the control signal easily and appropriately.

(174) The second function **M2** specifying a relation between the actual number of revolutions of the prime mover **6** and the second correction coefficient is stored in the storage unit **100a**, and the changer **100c** calculates the second correction coefficient by substituting the actual number of revolutions of the prime mover **6** detected by the second detector **103** into the second function **M2**. With this configuration, the changer **100c** is able to calculate the correction coefficient easily and accurately by using the function specifying the relation between the actual number of revolutions of the prime mover **6** and the second correction coefficient.

(175) The second function **M2** defines the second correction coefficient as a value equal to one

when the actual number of revolutions of the prime mover **6** is not less than the third threshold **Re2**, and defines the second correction coefficient in terms of value that is less than one when the actual number of revolutions of the prime mover **6** is less than the third threshold **Re2**. With this configuration, when the number of revolutions of the prime mover **6** is less than the third threshold **Re2** and is comparatively small, the changer **100c** is able to change the first line **La** in a direction in which the primary pressure outputted from the actuation valve **70** decreases. By this means, it is possible to reduce the output of the traveling pump **50** and thus prevent the engine from stalling.

(176) The controller **100** is switchable between a plurality of modes, and a plurality of second functions **M2** different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit **100a**, and the changer **100c** calculates the second correction coefficient based on, among the plurality of second functions **M2**, a second function **M2** that corresponds to a mode of the controller **100**. With this configuration, the controller is able to change the mode depending on the degree of priority, for example, “engine stall suppression first”, “operational feeling first”, “traveling power first”, and the like.

(177) The changer **100c** computes a third correction coefficient based on a product of the first correction coefficient and the second correction coefficient. The changer **100c** changes the control signal expressed by the first line **La** in a case where the value of the third correction coefficient is greater than one and does not change the control signal expressed by the first line **La** in a case where the value of the third correction coefficient is not greater than one. With this configuration, it is possible to prevent the changer **100c** from changing the first line **La** excessively in a direction in which the primary pressure outputted from the actuation valve **70** decreases. That is, it is possible to prevent the traveling power of the traveling motor **51** from being reduced excessively due to changing the first line **La** by the changer **100c** when the actual number of revolutions of the prime mover **6** is less than the third threshold **Re2**.

(178) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

## Claims

1. A working machine, comprising: a prime mover; a traveling pump to be operated by power of the prime mover and deliver a hydraulic fluid; a traveling motor, configured to transmit power to a drive shaft of a traveling device, to rotate using the hydraulic fluid delivered by the traveling pump; an operation valve to change pilot pressure of a pilot fluid outputted to the traveling pump in response to operation of an operation member; an actuation valve, provided upstream from the operation valve on a fluid passage, to operate based on a control signal and change primary pressure that is the pilot pressure of the pilot fluid supplied to the operation valve from the fluid passage; a controller configured or programmed to control opening of the actuation valve by outputting the control signal to the actuation valve; and a first detector to detect an actual number of revolutions of the traveling motor, wherein the controller includes a changer configured or programmed to change setting of the control signal such that the opening of the actuation valve increases as the actual number of revolutions of the traveling motor decreases.
2. The working machine according to claim 1, further comprising: a number-of-revolutions operation actuator configured for an operator to operate a target number of revolutions of the prime mover; a second detector to detect an actual number of revolutions of the prime mover; and a storage unit in which a first line and a second line are stored, the first line defining the control signal based on the actual number of revolutions of the prime mover in a case where a difference between the target number of revolutions of the prime mover and the actual number of revolutions of the prime mover is not less than a first threshold, the second line defining the control signal to be

greater than defined by the first line in a case where the difference is less than the first threshold, wherein the changer is configured or programmed to change the first line by changing the control signal expressed by the first line such that the opening of the actuation valve increases as the actual number of revolutions of the traveling motor decreases.

3. The working machine according to claim 2, wherein, based on a first correction coefficient defined in relation to the actual number of revolutions of the traveling motor, the changer is configured or programmed to change the control signal expressed by the first line.

4. The working machine according to claim 3, wherein, at least one first function specifying a relation between the actual number of revolutions of the traveling motor and the first correction coefficient is stored in the storage unit, and the changer is configured or programmed to calculate the first correction coefficient by substituting the actual number of revolutions of the traveling motor detected by the first detector into the at least one first function.

5. The working machine according to claim 4, wherein, the at least one first function defines the first correction coefficient in terms of value that is not less than one, such that the first correction coefficient corresponding to a second number of revolutions, which represents the actual number of revolutions of the traveling motor and is less than a first number of revolutions, is greater than the first correction coefficient corresponding to the first number of revolutions, which represents the actual number of revolutions of the traveling motor.

6. The working machine according to claim 5, wherein, the at least one first function has degrees of inclination which are different depending on whether the actual number of revolutions of the traveling motor is less than a second threshold thereof, and the degree of inclination of the at least one first function in a range where the actual number of revolutions of the traveling motor is equal to or less than the second threshold, is greater than the degree of inclination of the at least one first function in a range where the actual number of revolutions of the traveling motor is not less than the second threshold.

7. The working machine according to claim 4, wherein, the controller is operable to be switched among a plurality of modes, and as the at least one first function, a plurality of first functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit, and the changer is configured or programmed to calculate the first correction coefficient based on, among the plurality of first functions, a first function that corresponds to a mode of the controller.

8. The working machine according to claim 3, wherein, based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer is configured or programmed to change the control signal expressed by the first line.

9. The working machine according to claim 8, wherein, at least one second function specifying a relation between the actual number of revolutions of the prime mover and the second correction coefficient is stored in the storage unit, and the changer is configured or programmed to calculate the second correction coefficient by substituting the actual number of revolutions of the prime mover detected by the second detector into the at least one second function.

10. The working machine according to claim 9, wherein, the at least one second function defines the second correction coefficient as a value equal to one when the actual number of revolutions of the prime mover is not less than a third threshold, and defines the second correction coefficient in terms of value that is less than one when the actual number of revolutions of the prime mover is less than the third threshold.

11. The working machine according to claim 9, wherein, the controller is operable to be switched among a plurality of modes, and as the at least one second function, a plurality of second functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit, and the changer is configured or programmed to calculate the second correction coefficient based on, among the plurality of second functions, a second function

that corresponds to a mode of the controller.

12. The working machine according to claim 8, wherein, the changer is configured or programmed to compute a third correction coefficient based on a product of the first correction coefficient and the second correction coefficient, change the control signal expressed by the first line in a case where a value of the third correction coefficient is greater than one, and not change the control signal expressed by the first line in a case where the value of the third correction coefficient is not greater than one.

13. The working machine according to claim 5, wherein, the controller is operable to be switched among a plurality of modes, and as the at least one first function, a plurality of first functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit, and the changer is configured or programmed to calculate the first correction coefficient based on, among the plurality of first functions, a first function that corresponds to a mode of the controller.

14. The working machine according to claim 6, wherein, the controller is operable to be switched among a plurality of modes, and as the at least one first function, a plurality of first functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit, and the changer is configured or programmed to calculate the first correction coefficient based on, among the plurality of first functions, a first function that corresponds to a mode of the controller.

15. The working machine according to claim 4, wherein, based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer is configured or programmed to change the control signal expressed by the first line.

16. The working machine according to claim 5, wherein, based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer is configured or programmed to change the control signal expressed by the first line.

17. The working machine according to claim 6, wherein, based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer is configured or programmed to change the control signal expressed by the first line.

18. The working machine according to claim 7, wherein, based on a second correction coefficient defined in relation to the actual number of revolutions of the prime mover in addition to the first correction coefficient, the changer is configured or programmed to change the control signal expressed by the first line.

19. The working machine according to claim 10, wherein, the controller is operable to be switched among a plurality of modes, and as the at least one second function, a plurality of second functions different from one another in terms of inclination at least partially, to correspond to the plurality of modes, is stored in the storage unit, and the changer is configured or programmed to calculate the second correction coefficient based on, among the plurality of second functions, a second function that corresponds to a mode of the controller.

20. The working machine according to claim 9, wherein, the changer is configured or programmed to compute a third correction coefficient based on a product of the first correction coefficient and the second correction coefficient, change the control signal expressed by the first line in a case where a value of the third correction coefficient is greater than one, and not change the control signal expressed by the first line in a case where the value of the third correction coefficient is not greater than one.

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